

From the continuum to discrete substrates: 't Hooft and other turns

Conversation record, 2026-09-28 (user and Claude, with a from-memory answer by Claude Fable). All references below are quoted from memory and have not been verified; they are leads for the prior-art section of the fifth-postulate paper, and no result here is claimed.

Conversation with the user, 2026-09-28 night (not yet in any repo note; references from memory, unverified).

't Hooft (Fable's memory, high/medium confidence): renormalizability and $\beta < 0$ (Marseille 1972); only first two beta coefficients scheme-independent (limits "2 b0 per log" beyond one loop); renormalons (Erice late 1970s) = the atlas exponential ladder, threshold $2 b0 c = 4$ is the gluon-condensate renormalon; instantons $e^{-8 \pi^2/g^2}$, U(1) problem; 't Hooft loop $B(C)$ (NPB 138, 1978) = our mid-edge centre flip; twisted b.c. and e/m flux sectors (NPB 153, 1979), fractional charge $1/N \bmod 1$ ($1/3$ for SU(3), fits the trisection result); the round-12 non-based-cycle Aharonov-Bohm phase is the small-angle part of his flux-sector data (round-14 prompt now asks Astra to state the covariant reference sector by sector); abelian projection 1981; large N 1974. Cellular-automaton interpretation (1988 on, book 2016): $\hbar$ enters as a conversion constant tied to the time grain δt ; the floor comes from information loss (equivalence classes give H bounded below). This is the exact opposite of the user's thesis on the same RG ground: he supplies $\hbar$ from below; we ask whether consistent refinement forces it.

Candidate next Newton premise (proposed, not yet sent to Astra): take 't Hooft's information-loss/equivalence-class construction as the physical premise and ask whether it bounds the recorded Galileo comparison. Contact point: our thermodynamic-records note gave only $\eta \geq A_0 e^{-W/kBT}$, no floor. Put this to an Astra decision round after round 14.

User's remarks: "as soon as you really understand the RG then you don't understand any" (amused); read Wolfram in the eighties thinking "how will he reach a continuum limit", yet the Newton thesis says the continuum limit is problematic from the start, so "a bit bipolar". Claude's reply: same fault line from two sides; the atlas takes the limit and asks what survives.

Wolfram (Caltech ~1979, QCD phenomenology, Fox-Wolfram moments; cellular automata 1981-82; NKS 2002; hypergraph physics 2020). **Cvitanovic** (Kinoshita's group, QED g-2; birdtracks from colour factors, book 2008, useful notation for our SU(N) Brauer-Klimyk sums; chaos via the Feigenbaum-Cvitanovic RG fixed point, ChaosBook).

Other "Damascus" actors listed: divergence-driven: Heisenberg universal length (1938), Snyder quantized spacetime (1947), Yukawa elementary domains, Dirac's lifelong rejection of renormalization. RG masters going deterministic/discrete: T. D. Lee discrete time (1983), Adler trace dynamics (2004), H. B. Nielsen random dynamics, Feynman checkerboard and "Simulating physics with computers" (1982). Geometry to combinatorics: Regge, Penrose spin networks, Finkelstein space-time code (1969), Sorkin causal sets, Ambjorn-Loll, Wheeler "it from bit", von Weizsacker ur-alternatives. **Closest to the Newton thesis:** Gisin and Del Santo (~2019-21), finite information makes classical mechanics indeterministic (denial of the *velocitas ultima* from quantum information) but give no $\hbar$; the fifth-postulate paper should cite and answer them. Proposed paper sentence: field theorists went discrete to escape infinities; RG masters to explain the quantum; Gisin/Del Santo to break determinism; the user's thesis keeps the continuum and shows consistent refinement demands the constant Newton's limit dropped.

Related: STATE, fifth postulate, stochastic route, thermodynamic records.

Consequence for STATE

None yet. The candidate Newton premise (information loss as in 't Hooft's equivalence classes) goes to an Astra decision round after round 14.